Polynomials

Integers and polynomials share one arithmetic. Both can be divided with a remainder, and from that single tool everything else follows: greatest common divisors, Bézout identities, unique factorization into primes or irreducibles, and the Chinese Remainder Theorem. This chapter develops that arithmetic for $F[x]$ in full, because from Chapter 9 on the structure of an operator is read off from polynomials: which polynomials kill it, how they factor, and which of their factors are coprime.

Along the way we keep two promises made in Chapter 0: we prove division with remainder, stated there only for orientation, and the Fundamental Theorem of Algebra, that every nonconstant complex polynomial has a root. We also prove that polynomials factor uniquely into irreducibles, just as integers factor uniquely into primes. The chapter ends by plugging an operator into a polynomial, where the arithmetic of $F[x]$ turns into a splitting of the space into pieces.

What you need. From Chapter 0, induction and well-ordering (Theorem 0.2.10, Theorem 0.2.11), divisibility of integers and Bézout's lemma (Definition 0.2.2, Lemma 0.7.2), the fields $\mathbb{F}_p$ (Theorem 0.7.3), polynomials, degree and evaluation (Definition 0.8.1, Definition 0.8.3, Theorem 0.8.3, Theorem 0.8.5), and complex numbers (Theorem 0.6.6). From Chapter 1, the standard bases of the polynomial spaces and their dimensions (Example (Standard Bases), Example (Dimensions of the Running Examples)), counting instead of checking (Theorem 1.6.4, Theorem 1.6.3), and direct sums (Definition 1.7.2, Theorem 1.7.4), which is what the kernel splitting lemma of §07 produces. From Chapter 2, polynomials of an operator (Definition 2.4.2). From Chapter 3, the factor theorem and root bound for linear factors (Lemma 3.7.1, Lemma 3.7.2) and interpolation (Theorem 3.7.3). From Chapter 4, quotient spaces and the First Isomorphism Theorem (Definition 4.4.3, Theorem 4.4.7). From Chapter 5, dual bases (Theorem 5.1.3).

6.-1.1 Roadmap

- **Division with remainder.** Long division always works over a field, with a unique quotient and remainder; the remainder and factor theorems in general.
- **Ideals and greatest common divisors.** Every ideal of $F[x]$ is generated by one polynomial; gcds, Bézout identities, and the Euclidean algorithm.
- **Irreducible polynomials and unique factorization.** The primes of $F[x]$, and why factorization into them is unique.
- **Roots, multiplicity, and derivatives.** Repeated roots detected by the formal derivative, and Taylor expansion of a polynomial.
- **The Fundamental Theorem of Algebra.** A complete proof, and what it says about real polynomials.
- **Interpolation and the Chinese Remainder Theorem.** The Lagrange formula, solving simultaneous congruences, and the quotient-space picture behind both.
- **Polynomials of an operator.** Every operator on a finite-dimensional space satisfies a polynomial, and coprime factors split the space.

6.-1.2 Named moves

- **Divide and look at the remainder.** Most statements about divisibility reduce to one division.
- **The smallest-degree element generates.** In an ideal, a nonzero element of least degree divides every other element.
- **Evaluate at a root.** Plugging a root of the divisor into $f = qg + r$ kills qg and reads off

information about the remainder.

- **Compare degrees.** Uniqueness of quotients, factorizations, and interpolants usually comes from a degree count.
- **Bézout gives projections.** When $\gcd(p, q) = 1$, write $1 = ap + bq$; evaluated at an operator, the two terms split the space.